

Sertaç Akalın

Computer Engineer — AI Engineer

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Summary

Computer Engineer with a research-oriented profile in computer vision, applied AI, and backend systems. I am most comfortable with open-ended technical problems that require building the evidence as well as the software: collecting data, defining an evaluation method, running experiments, measuring failure cases, and iterating on the system. For my traffic-violation research, I independently collected and annotated a custom Istanbul traffic dataset, trained and evaluated a YOLOv8-based detection pipeline, combined it with tracking and spatial-temporal decision logic, and validated the result on real traffic videos. The work became a first-author paper accepted at IDAP 2026 and is being extended toward a journal manuscript. Alongside research, I have hands-on backend experience with Python/FastAPI and Java/Spring Boot in production-oriented engineering environments.

Publications & Research

International Conference Papers (IEEE Xplore)

- [1] M. F. Şen, H. Gümüşkaya, and **S. Akalın**, “A deployable dual-mode vision framework for adaptive intersection control and traffic analytics,” in *Proc. 10th Int. Artificial Intelligence and Data Processing Symp. (IDAP 2026)*, Istanbul, Türkiye, Sep. 5–6, 2026. **Awarded Best Applied Research Paper, IDAP 2026.**
- [2] **S. Akalın** and H. Gümüşkaya, “Computer vision-based detection of vehicles violating hatched road areas in Istanbul traffic,” in *Proc. 10th Int. Artificial Intelligence and Data Processing Symp. (IDAP 2026)*, Istanbul, Türkiye, Sep. 5–6, 2026.

Ongoing Project and Manuscript Work

- [3] **S. Akalın** and H. Gümüşkaya, extended version of [2], in preparation for submission to a Q1/Q2-indexed journal.
- [4] **S. Akalın**, H. Gümüşkaya, and M. F. Şen, extended version of [1], in preparation for submission to a Q1/Q2-indexed journal.

Technical Skills

Programming: Python, Java

AI / Computer Vision: PyTorch, YOLOv8, ByteTrack, OpenCV, object detection, object tracking, model training and evaluation

Data & Research Workflow: dataset collection and annotation, Roboflow, quantitative evaluation, real-video field testing

Backend & Databases: FastAPI, Spring Boot, REST APIs, PostgreSQL, MySQL, JWT authentication

Tools & Deployment: Git, Docker, CI/CD (GitHub Actions), Google Cloud Run, Postman, Swagger

Experience

Software Engineering Intern — FEV Türkiye Jan 2026 – Jun 2026 (6 months)

- Developed Python/FastAPI backend services for SOVD (Service-Oriented Vehicle Diagnostics), an external project delivered to Jaguar Land Rover / Tata.
- Designed and implemented REST-based diagnostic service endpoints used in vehicle-diagnostics workflows.
- Worked daily with engineering teams across Germany and India in an Agile/Scrum environment, including stand-ups, code-oriented collaboration, and enterprise development processes.

Java Backend Developer Intern — CodeFirst Jan 2025 – Apr 2025 (4 months)

- Developed RESTful APIs for a Spring Boot microservice system using a layered Controller–Service–Repository architecture, JWT-based authentication, and Spring Data JPA database operations.
- Used Git, Docker, and Maven for version control, environment management, and builds; tested and documented APIs with Postman and Swagger.

Projects

Hatched Area Violation Detection — GitHub *YOLOv8, ByteTrack, OpenCV, Shapely, PaddleOCR — accepted, IDAP’26*

- Built a computer-vision system for detecting vehicles that enter hatched road areas in fixed-camera Istanbul traffic video, from data preparation through field evaluation and interface delivery.
- Independently collected and annotated the research dataset: 3,897 source frames and 9,353 exported images. Fine-tuned YOLOv8m and reached 0.773 mAP50.
- Designed a spatial-temporal state machine using polygon geometry and multi-frame confirmation to reduce false violation decisions caused by detector noise.
- Evaluated the full system on 11 real traffic videos, reaching 0.889 precision, recall, and F1; added severity scoring, evidence logging, and license-plate recognition in a Gradio interface.
- Publicly released the dataset and code, and used the study as the basis of a first-author IDAP 2026 paper and an extended journal manuscript in preparation.

Palmystra — AI-Powered Mobile App *FastAPI, LangChain, RAG, Computer Vision, SwiftUI*

- Built and shipped a full-stack iOS application combining computer vision with a LangChain-based RAG pipeline to generate personalized readings from a palm photo.
- Implemented authentication, rate limiting, and automated tests for the backend; containerized the service with Docker and deployed it to Google Cloud Run.

Education

İstanbul Arel University — B.Sc. in Computer Engineering (English)	2022–2026
Associate Degree — Web Design and Coding	2020–2022

Languages: Turkish (native), English (B2) • **Military Service:** Completed